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Llama-2-hf non stopping token generation. #24994

[Closed](#)[4 tasks](#)

francescobodria opened this issue on Jul 21, 2023 · 8 comments

francescobodria commented on Jul 21, 2023

System Info

- transformers version: 4.31.0
- Platform: Linux-5.15.0-1035-azure-x86_64-with-glibc2.31
- Python version: 3.9.16
- Huggingface_hub version: 0.16.4
- Safetensors version: 0.3.1
- Accelerate version: 0.22.0.dev0
- Accelerate config: not found
- PyTorch version (GPU?): 1.13.1+cu116 (True)
- Tensorflow version (GPU?): not installed (NA)
- Flax version (CPU?/GPU?/TPU?): not installed (NA)
- Jax version: not installed
- JaxLib version: not installed
- Using GPU in script?:
- Using distributed or parallel set-up in script?:

Who can help?

No response

Information

- ☐ The official example scripts
- ☐ My own modified scripts

Tasks

- ☐ An officially supported task in the `examples` folder (such as GLUE/SQuAD, ...)
- ☐ My own task or dataset (give details below)

Reproduction

```
from transformers import AutoModelForCausalLM, AutoTokenizer, pipeline
import torch
```

```
model_id = "/home/modelweights/llama2-hf-7b"
tokenizer = AutoTokenizer.from_pretrained(model_id)
model = AutoModelForCausalLM.from_pretrained(
    model_id,
    device_map='auto',
    load_in_8bit=True,
)
```

```
pipe = pipeline(
    "text-generation",
    model=model,
    tokenizer=tokenizer,
    do_sample=True,
    top_k=50,
    top_p=0.95,
    num_return_sequences=1,
    eos_token_id=tokenizer.eos_token_id,
```



Hi,

My Llama 2 model is not generating the stopping tokens.

For example the reply of the question Hello there! How are you doing? is:

Result: Hello there! How are you doing? I hope you are doing well. I am doing well. I am doing well. I am doing well. I am doing well. I am doing well. I am doing well. I am doing well. I am doing well. I am doing well. I am doing well. I am doing w

The reply only stops when the max token criteria is met

What am I doing wrong?

sgugger commented on Jul 21, 2023

Collaborator

cc @gante and @ArthurZucker

ArthurZucker commented on Jul 21, 2023 • edited ▾

Collaborator

Hey! This is expected, the llama model kind of rarely generates the `eos_token` . It was the same with Llama 1, and if you run your script with the original llama, you will get the same output:

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```
import fire
```

```
from llama import Llama
```

```
def main(
    ckpt_dir: str,
    tokenizer_path: str,
    temperature: float = 0.6,
    top_p: float = 0.9,
    max_seq_len: int = 512,
    max_gen_len: int = 256,
    max_batch_size: int = 8,
):
    generator = Llama.build(
        ckpt_dir=ckpt_dir,
        tokenizer_path=tokenizer_path,
        max_seq_len=max_seq_len,
        max_batch_size=max_batch_size,
    )
```

```
results = generator.text_completion(  
    ['Hello there! How are you doing?'],  
    max_gen_len=max_gen_len,  
    temperature=temperature,  
    top_p=top_p,  
)
```

```
print('Hello there! How are you doing?')
print(f"> {results['generation']}")
print("\n=====\\n")
```

```
if __name__ == "__main__":
    fire.Fire(main)
```

and

will produce:

> I hope you are doing well. I am doing well. I am happy to be back here. I have been away for a while. I have been busy with my stu



Note that the setting you are providing (`temperature = 0.1` etc) can influence on the generation.



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AnishAlapattu-GEP commented on Jul 25, 2023

So what are the best practices currently known to reduce this random ramble?

ArthurZucker commented on Jul 25, 2023 • edited

Collaborator

You should probably ask this question on [the forum](#), using the default parameters should already help (`temperature=0.9`), you can try to use `LogitsProcessor` for length penalty to reduce potential hallucination. You can also change the sampling strategies, use `top_k` , `contrastive search` etc etc. [@gante](#) will have better solution!

gante commented on Jul 25, 2023 • edited

Member

[@AnishAlapattu-GEP](#) [@ArthurZucker](#)

This is actually a hard problem to solve, and I have yet to see a solution that generalizes well! A few things that can be tried, ranked by implementation complexity:

1. In the prompt, mention that you want a short output (be specific if you can, like "Reply in 3 sentences or less");
2. Add a custom logits processor that increases the score of the eos token according to some rule (e.g. scaling with the generated length) *EDIT* it is implemented in [this logits processor](#)!
3. Generate text in excess and have a post-processing step to crop unwanted text (e.g. based on the conditional probability of each sequence -- when the model starts rambling, I suspect there is a significant drop in the probability of the sentence given the past sentences)
4. Fine-tune the model 🙄



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AnishAlapattu-GEP commented on Jul 25, 2023

Thanks [@ArthurZucker](#) and [@gante](#)!



[francescobodria](#) closed this as [completed](#) on Aug 4, 2023

eryk-mazus commented on Sep 1, 2023

The issue stems from using bare Llama-2 model, instead of `-chat` version, which is fine-tuned to follow instructions.

Bare llama-2 model is trained to complete text, so if you include the beginning of the conversation in the prompt, you should expect the rest of the conversation to be predicted by such model. In contrast, `-chat` models are trained to be more aligned or to follow the human instructions. This idea is nicely presented by openai in InstructGPT [article](#)

Completion GPT-3

Explain the theory of gravity to a 6 year old.

Explain the theory of relativity to a 6 year old in a few sentences.

Explain the big bang theory to a 6 year old.

Explain evolution to a 6 year old.

InstructGPT

People went to the moon, and they took pictures of what they saw, and sent them back to the earth so we could all see them.



gante commented on Sep 4, 2023 • edited ▾

Member

👍 Precisely

A thing I forgot to mention in my answer above (which I've edited in case someone stumbles upon it): we have a way to add a soft cap on the generated text length. e.g. If you want your output to be about 100 tokens unless the remaining tokens are really important for the answer, you can do it through [this logits processor](#).

🔗 dongwang218 mentioned this issue on Sep 8, 2023

LLaMA can't generate eos token meta-llama/llama#321

🔒 Closed

Assignees

No one assigned

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

No branches or pull requests

6 participants

